🔷 16.9 Global System Closure and Unified Guarantees

16.9.1 Unified System State

$$\mathcal{S}_k = (\Psi_k, P_k, PAC_k, L_k, D_k, u_k)$$

$$\mathcal{S}_k \in \mathcal{X}_{\text{safe}}$$

where $\mathcal{X}_{\text{safe}}$ denotes the admissible safe state space.

16.9.2 Global Evolution Law

$$\mathcal{S}{k+1} = F{\text{global}}(\mathcal{S}_k)$$

with:

$$F_{\text{global}} = F_{\text{dyn}} \circ F_{\text{prob}} \circ F_{\text{PAC}} \circ F_{\text{graph}} \circ F_{\text{decision}} \circ F_{\text{control}}$$

Thus:

the system evolves through a fully integrated composition of all subsystems.

16.9.3 Closed-Loop Structure

$$\Psi \rightarrow P \rightarrow PAC \rightarrow L \rightarrow D \rightarrow u \rightarrow \Psi$$

Thus:

the system operates with no external dependency.

16.9.4 Feasibility Invariance

$$\mathcal{S}_k \in \mathcal{X}_{\text{safe}} \;\Rightarrow\; \mathcal{S}_{k+1} \in \mathcal{X}_{\text{safe}}$$

Thus:

the system remains within the safe domain at all times.

16.9.5 Global Lyapunov Stability

$$
\mathcal{L}(\mathcal{S}) =
\mathcal{E}(\Psi) + KL(P \| P_0)
$$

$$
\frac{d\mathcal{L}}{dt}
\le
- c \|\nabla \mathcal{E}\|^2
+ C \sigma^2
$$

Thus:

global stability is preserved under stochastic perturbations.

16.9.6 Robustness

Assuming:

- bounded noise
- bounded delay
- bounded disturbances

then:

$$
\|\mathcal{S}_k\| \le C
$$

Thus:

the system remains globally bounded.

16.9.7 Convergence

$$\lim_{k \to \infty} \mathcal{S}_k = \mathcal{S}^*$$

Thus:

the system converges to a steady-state solution.

16.9.8 Optimality

$$u_k = \arg\max_{u \in \mathcal{U}_{\text{feasible}}} D_u(k)$$

Thus:

the system follows an optimal decision policy.

16.9.9 Adaptivity

- PAC evolves dynamically
- network topology adapts
- control law updates in real time

Thus:

the system is fully self-adaptive.

16.9.10 Real-Time Guarantee

T_{\text{total}} \le T_{\text{critical}}

Thus:

the system satisfies real-time constraints.

16.9.11 Safety Closure
- state constraints enforced
- fail-safe mechanisms active
- bounded trajectories ensured

Thus:
\mathcal{S}_k \in \mathcal{X}_{\text{safe}}

16.9.12 Multi-Agent Consistency

$$\lambda_2(L) > 0$$

Thus:

the system achieves consensus across agents.

16.9.13 Deterministic Limit

$$\sigma \to 0
\;\Rightarrow\;
\mathcal{S}_k \to \mathcal{S}^*$$

Thus:

the system reduces to a deterministic optimal solution.

Final Master Statement

\boxed{

\text{SEKHEM-TSDD defines a fully closed, stochastic, triadic decision–control system with guaranteed stability, safety, optimality, and adaptivity.}
}